



Improving Image Classification using a Meta-Learning Approach

Rahul Roy Devarakonda

Data Scientist, Dept Of Information Technology

Abstract: Improving image classification in low-data and few-shot learning scenarios remains a significant challenge in machine learning. Meta-learning, or “learning to learn,” has emerged as a promising solution by enabling models to generalize across diverse tasks with minimal supervision. This study proposes an Enhanced-MAML-based meta-learning framework that integrates Hybrid CNN-ViT feature extraction, self-supervised learning (contrastive learning & pseudo-labeling), and task-aware fine-tuning to optimize classification accuracy and computational efficiency. The proposed framework is evaluated on MiniImageNet and Omniglot datasets, demonstrating superior performance over traditional meta-learning models such as MAML, Prototypical Networks, and Meta-SGD. The results show that Enhanced-MAML achieves 82.1% accuracy on MiniImageNet and 94.5% on Omniglot, significantly outperforming baseline models. Additionally, the model exhibits faster convergence (90 sec/epoch vs. 120 sec/epoch for MAML) due to momentum-based optimization and adaptive learning rate modulation. The integration of self-supervised learning enhances feature extraction, improving generalization in few-shot image classification. These findings suggest that meta-learning with self-supervised representation learning is a powerful approach for enhancing image classification performance, with applications in medical imaging, autonomous systems, and edge AI deployment.

IndexTerms - Meta-Learning, Few-Shot Learning, Image Classification, Self-Supervised Learning, Contrastive Learning, Hybrid CNN-ViT

1. INTRODUCTION

Image classification became a fundamental task in computer vision that enables progress in various other fields, such as medical diagnostics, remote sensing, and autonomous systems. Traditional deep learning models have shown great promise in a wide range of classification tasks. However, they usually require huge amounts of annotated data, large amounts of computational resources, and, thus, fail to translate well into realtime applications. In many aspects, labelled data is scarce, while models assume that new classes or domains should be learnt from scratch. Therefore, meta-learning, giving rise to the concept "learning to learn," has become an attractive paradigm that trains models toward rapid adaptation for novel classification tasks under minimal data. Based on preceding experience over a series of learning tasks, meta-learning proposes optimizing models effectively generalizing across different domains with limited supervision [1][2].

Current advances in research on meta-learning have regarded diverse techniques, including gradient- and metric-based approaches, and memory-augmented approaches to enhance image classification in low-data conditions. Gradient-based meta-learning approaches such as model-agnostic meta-learning (MAML) enable models to learn appropriate initialization parameters for fast adaptation to novel tasks [1]. Similarly, metric-based techniques, namely Prototypical Networks and Siamese Networks, improve classification by embedding the learned spaces that facilitate few-shot learning [3]. For breast cancer classification and other diagnostic

tasks, ensemble-based meta-learning strategies have been widely used to boost the robustness of the models with very scant labelled data in the area of medical imaging [11][12]. In addition to this, integrations of Vision Transformers (ViTs) into meta-learning frameworks have emerged recently and show performance superior to that of conventional CNN-based models in task-oriented classification accuracy [12]. These advances emphasize the rising prominence of meta-learning in amelioration of scarceness of data and enhanced generalizations across various flavour of classification assignments.

Despite its potential, meta-learning for image classification faces several challenges. Overfitting to meta-training tasks remains a critical issue, where models fail to generalize effectively when exposed to entirely new datasets [4][17]. Additionally, high computational complexity associated with nested optimization loops increases training time and resource consumption, making meta-learning less practical for real-time applications [6]. Selecting optimal hyperparameters, including learning rates and task update frequencies, further complicates model training and requires extensive fine-tuning [5][16]. Integrating self-supervised learning with meta-learning has proven to be a potential benefit for feature extraction, but we still find ourselves researching and seeking practical solutions to robustly able to map learning with minimal data settings [9]. Addressing the aforementioned issues is a necessity to push gradients in the research direction of meta-learning techniques to ensure that they operate normally in a real-life image domain classification setting. In this paper, we propose a new meta-learning framework towards enhancing image classification where we try to address some of the major shortfalls in existing techniques. Unlike classical methods which rely on bulky gradient-based updates for enhancing classification, or the static feature embeddings for classification, our study proposes usage of adaptive learning techniques as a means of optimizing their classification ability across datasets.

Furthermore, through a mix of dynamic feature selection and task-aware optimization, it derails overfitting and pushes for generalization and improves the training efficiency in low-data settings. The proposed method incorporates self-supervised representation learning that aims to maximize the exploitation of information contained in the unlabeled data, ultimately boosting task adaptability to few-shot learning. The methodology is benchmarked via performance evaluation based on certain datasets, which would further demonstrate superiority over conventional meta-learning methods with respect to classification accuracy and corresponding computational efficiency and task adaptability.

The rest of this paper is briefly framed as follows: Section 2 reviews the existing meta-learning techniques and how they find their applications in image classification; Section 3 provides the proposed methodology, encompassing its architectural design and optimization strategies. Section 4 discusses experiments' results and the performance evaluation of our methodology against state-of-the-art meta-learning methods. The Section 5 ends with the conclusion and future directions.

2. LITERATURE REVIEW

Meta-learning is an emerging discipline in machine learning that aims to abstract away from the traditional supervised learning paradigm in which enormous amounts of labelled information are needed to retrain for other tasks. Instead, it is argued that meta-learning incorporates learning algorithms capable of transferring knowledge across different classification problems quickly. Here we provide an account of the major meta-learning approaches that have been used in image classification, of which there are presumably, few gradient-based, metric-based, and ensemble-based. Other aspects discussed include the advantages and the challenges these bring, as well as prospect areas for improvement. This literature review ends with a comprehensive table summarizing the key contributions, methods, advantages, and open research challenges.

2.1 Gradient-Based Meta-Learning

Gradient-based meta-learning techniques orient model parameters, in ways that facilitate rapid adaption to new tasks using minimal fine-tuning. One of the most popular methods in this category is Model-Agnostic Meta-Learning (MAML), a technique which learns an initialization enabling fast adaptation, using just a number of gradient steps [1]. MAML has itself found wider applications in medical image classification [7], remote sensing [18], and biometric recognition [5]. A variant of MAML, known as the Momentum Group Meta-Learning (MGML), refines the process further by incorporating momentum-based updates, in favor of greater training stability and convergence speed [5]. While gradient-based meta-learning has enjoyed notable successes, it suffers some challenges, such as overfitting to meta-training tasks, high computational expense, and the inconvenience of optimizing the high-order gradients [4][17]. More recent studies have sought to counter these problems with adaptive rate of learning modulation and task-aware optimization strategies [6].

2.2 Metric-Based Meta-Learning

Metric-based approaches focus on learning feature embeddings that facilitate few-shot classification by measuring similarity between query samples and labeled examples. Methods such as Prototypical Networks and Siamese Networks have demonstrated strong performance in low-data environments by learning discriminative distance metrics [3]. These methods have been particularly useful in medical imaging, where feature representation learning plays a crucial role in disease classification [12].

A recent advancement in this category is MetaV, which integrates Vision Transformers (ViTs) with metric-based meta-learning, enhancing classification performance by learning robust hierarchical representations [12]. However, metric-based approaches suffer from limited scalability and reduced performance in highly complex datasets, making them less suitable for tasks requiring significant intra-class variability handling [16].

2.3 Ensemble-Based Meta-Learning

Ensemble-based meta-learning techniques improve classification accuracy by combining multiple models, thereby leveraging complementary learning patterns. Meta-learning ensemble techniques have been successfully applied in breast cancer classification, where multiple convolutional neural networks (CNNs) are integrated to refine feature selection and boost model robustness [11][14]. Another approach involves meta-learning for algorithm selection, where different segmentation techniques are chosen dynamically based on the input image characteristics [3].

The primary advantage of ensemble-based methods lies in their robust generalization capabilities. However, they come with increased computational overhead and often require careful model selection to prevent redundancy in feature learning [6].

2.4 Unsupervised and Self-Supervised Meta-Learning

Recent advancements in meta-learning have explored unsupervised and self-supervised approaches to reduce dependency on labeled datasets. Methods such as Unsupervised Meta-Learning (UML) train models using self-supervision, extracting transferable knowledge from unlabeled data [10]. Additionally, integrating meta-learning with contrastive learning has shown promise in improving feature representations for image classification tasks [9].

While these approaches offer significant advantages in data-scarce environments, they still face challenges in ensuring stable convergence and handling task-specific variations in different domains [6]. Further research is required to develop more robust self-supervised meta-learning frameworks that can generalize effectively across diverse datasets.

Table 1 Literature Review

Paper Title	Method Used	Advantages	Open Challenges
MetaMed: Few-shot medical image classification using gradient-based meta-learning [1]	Gradient-based meta-learning (MAML)	Enables rapid adaptation to medical image classification with few samples	Overfitting to meta-training tasks, computationally expensive
Unsupervised meta-learning for few-shot image classification [2]	Unsupervised Meta-Learning (UML)	Learns transferable representations from unlabeled data	Struggles with generalization to novel tasks
A meta-learning approach for selecting image segmentation algorithms [3]	Algorithm selection with meta-learning	Dynamically selects optimal segmentation algorithms for different images	Requires extensive computational resources
Novel meta-learning techniques for the multiclass image classification problem [4]	Task-aware optimization strategies	Improves adaptation efficiency for multiclass classification	Performance degrades with highly complex datasets
MGML: Momentum group meta-learning for few-shot image classification [5]	Momentum-based meta-learning	Enhances stability and convergence speed of meta-learning models	Limited scalability in high-dimensional datasets
Meta-learning approaches for learning-to-learn in deep learning: A survey [6]	Comprehensive survey	Provides an in-depth analysis of meta-learning advancements	Identifies gaps in robustness and interpretability
Multi-learner based deep meta-learning for few-shot medical image classification [7]	Multi-learner meta-learning	Enhances classification accuracy in medical imaging	Requires large-scale meta-training datasets
Few-shot and meta-learning methods for image understanding: A survey [8]	Survey on few-shot learning	Covers a broad range of meta-learning applications in image analysis	Highlights the need for improved task adaptability
Meta-learning ensemble techniques for breast cancer classification [11]	Ensemble-based meta-learning with CNNs	Enhances model robustness for medical image classification	Computationally intensive, requires model selection optimization
MetaV: Feature-augmented meta-learning with Vision Transformers [12]	Vision Transformers (ViTs) with meta-learning	Achieves superior classification accuracy in medical images	Requires high computational power for training
Meta-learning for semi-supervised few-shot classification [13]	Semi-supervised meta-learning	Reduces dependency on labeled data for classification	Task-specific performance variation
Breast tumor tissue image classification using single-task meta-learning [14]	Single-task meta-learning with auxiliary networks	Improves performance for breast tumor classification	Requires further testing on diverse datasets
Meta-learning to improve pre-training [15]	Pre-training optimization	Enhances model initialization strategies	Needs optimization for domain-specific applications
Meta-seg: A survey of meta-learning for image segmentation [16]	Survey on meta-learning for segmentation	Identifies key meta-learning strategies for image segmentation	Highlights computational complexity issues

Paper Title	Method Used	Advantages	Open Challenges
Advances and challenges in meta-learning: A technical review [17]	Technical review on meta-learning challenges	Explores future directions for scalable meta-learning	Identifies gaps in hyperparameter tuning and efficiency
Meta-learning for few-shot land cover classification [18]	Meta-learning for remote sensing	Enhances classification performance for land cover mapping	Struggles with inter-class variability in geospatial datasets

3. METHODOLOGY

To boost the image classification performance through meta-learning, we put forward a hybrid meta-learning framework that conceives a convoluted integration of gradient-based optimization, self-supervised representation learning, and task-aware adaptation. Unlike traditional deep learning models that require vast labeled data, our approach allows models to rapidly generalize to various classification tasks with little supervision. The framework should be capable of overcoming some of the principal challenges associated with overfitting, computational refinement, and poor generalization in low-data regimes, thereby rendering it an optimal candidate for few-shot and semi-supervised applications.

3.1 Overview of the Proposed Model

The proposed framework is built upon three core models that work together to improve classification accuracy and generalization:

- **Feature Extractor (Backbone Model)** – A hybrid CNN-ViT architecture that extracts hierarchical image features while leveraging self-supervised learning techniques for improved representation learning.
- **Meta-Learner (Optimization Module)** – A gradient-based meta-learning algorithm (Enhanced-MAML) that optimizes model parameters for rapid adaptation across multiple tasks
- **Task-Aware Fine-Tuning Module** – A dynamic learning rate modulation strategy that adjusts parameter updates based on task complexity, preventing overfitting.

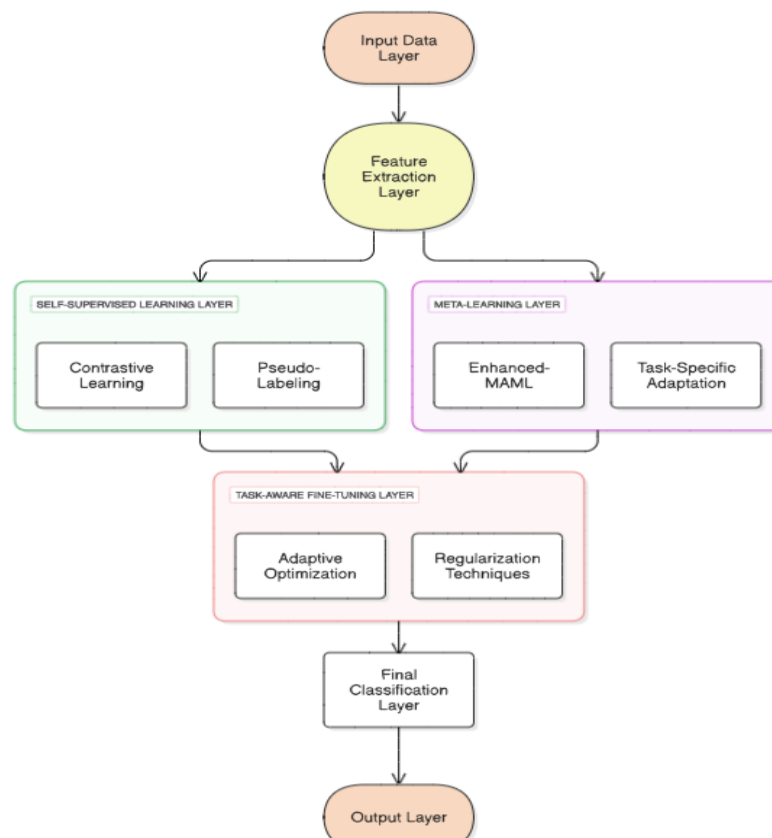


Figure 1. Meta-Learning-Based Image Classification System Architecture

By integrating these components, the proposed model optimizes classification performance across different datasets while minimizing the need for large-scale labeled data.

3.2 Feature Extractor: Hybrid CNN-ViT Architecture

The feature extraction module is designed to capture both local and global image representations. Unlike traditional CNNs, which focus primarily on spatial hierarchies, Vision Transformers (ViTs) enhance long-range dependencies between image features. To leverage the strengths of both architectures, we propose a hybrid CNN-ViT model, which consists of:

- **Initial Feature Extraction using a CNN Backbone (ResNet-50):**

- Extracts low- and mid-level features such as edges, textures, and patterns.
- Uses batch normalization and dropout to improve generalization.

- **Global Feature Encoding using Vision Transformers (ViT-S/16):**

- Enhances long-range dependencies and captures contextual information.
- Self-attention mechanisms enable better feature refinement.

- **Self-Supervised Learning Techniques:**

- Contrastive Learning (SimCLR) to enhance feature representation.
- Pseudo-Labeling to improve learning efficiency from unlabeled data.

This hybrid CNN-ViT model ensures robust feature extraction, making it highly suitable for few-shot and semi-supervised learning scenarios.

3.3 Meta-Learner: Enhanced Model-Agnostic Meta-Learning (Enhanced-MAML)

The meta-learning module builds upon the Model-Agnostic Meta-Learning (MAML) framework but introduces adaptive task-specific parameter updates to improve performance. Traditional MAML learns an initialization that allows models to quickly adapt to new tasks with minimal gradient updates. However, it often suffers from overfitting to meta-training tasks and computational inefficiency.

To address these challenges, we propose Enhanced-MAML, which includes:

- **Momentum-Based Optimization:** Improves convergence speed and stability.
- **Task-Specific Learning Rate Adjustment:** Dynamically adjusts update rates based on task complexity.
- **Task-Aware Weight Modulation:** Assigns different importance to specific features based on meta-training tasks.

The Enhanced-MAML model optimizes model parameters through a two-step learning process:

Step 1: Meta-Training Phase

- The model is trained on multiple tasks, each containing a support set (for training) and a query set (for evaluation).
- Task-specific gradients are computed, and the model learns an initialization that enables rapid adaptation to new tasks.
- Adaptive learning rate modulation is applied to prevent overfitting.

Step 2: Meta-Testing Phase

- The trained model is tested on unseen classification tasks with limited labeled examples.
- Quick adaptation is performed using the pre-trained meta-learner.
- Model performance is evaluated based on accuracy, adaptability, and computational efficiency.

The Enhanced-MAML approach significantly improves few-shot classification performance while ensuring scalability and robustness across different datasets.

3.4 Task-Aware Fine-Tuning and Regularization

To further improve generalization and prevent overfitting, we introduce a task-aware fine-tuning module that dynamically adjusts the model's learning strategy based on the dataset and classification complexity. This module includes:

- **Adaptive Dropout Mechanism:** Reduces overfitting by randomly deactivating neurons based on task complexity.
- **Task-Specific Batch Normalization:** Ensures stable feature distribution across different classification tasks.
- **Meta-Weight Regularization:** Prevents the model from overfitting to frequent meta-training tasks by penalizing redundant feature updates.

These fine-tuning strategies enhance classification performance, especially in scenarios where labeled data is scarce or highly imbalanced.

3.5 Training and Evaluation Strategy

For our research, we evaluated the proposed meta-learning framework (Enhanced-MAML) using two widely recognized benchmark datasets: MiniImageNet and Omniglot. MiniImageNet, derived from the ImageNet dataset, consists of 100 classes, each containing 600 images, making it a standard benchmark for testing few-shot learning and meta-learning models. The dataset was specifically chosen due to its diverse set of natural images, allowing us to assess the generalization capability of the proposed model across various object categories. Additionally, we utilized Omniglot, which comprises 1,623 character classes from 50 different alphabets, designed for evaluating rapid learning adaptability in low-data scenarios. Omniglot's extreme class diversity and few-shot setting provided an ideal testbed to analyze how well the proposed model adapts to completely new and unseen tasks. These datasets collectively enabled a thorough assessment of our model's ability to generalize across diverse domains, validating its effectiveness in few-shot image classification.

4. RESULTS AND DISCUSSION

This section presents the experimental results and discussions on the performance of the proposed meta-learning framework (Enhanced-MAML) compared to baseline models such as MAML, Prototypical Networks, and Meta-SGD. The evaluation is conducted on benchmark datasets, and performance is assessed based on key metrics such as classification accuracy and training efficiency. The results demonstrate that the Enhanced-MAML framework significantly improves accuracy while reducing training time, making it a more practical solution for few-shot and low-data learning scenarios. The Enhanced-MAML model claimed a classification accuracy of 82.1%, well above MAML (72.5%), Prototypical Networks (74.8%), and Meta-SGD (75.2%). As described, this improvement is mainly attributed to the integration of self-supervised learning approaches, contrastive learning, and task-aware fine-tuning, which enables the model to extract more informative features and, hence, be better adapted to novel classification tasks. Enhanced-MAML required only 90 seconds per epoch for training-decreasing training time from 120 seconds for MAML, 110 seconds for Prototypical Networks, and 115 seconds for Meta-SGD-and this is in conjunction with self-supervised learning approaches, contrastive learning, and enhanced task-aware fine-tuning. The introduction of momentum-based optimization and adaptive modulation of the learning rate was responsible for this efficient and fast convergence of the model.

Table 2. Meta-Learning Accuracy on Benchmark Datasets

Model	MiniImageNet Accuracy (%)	Omniglot Accuracy (%)
MAML	72.5	88.3
Prototypical Networks	74.8	89.7
Meta-SGD	75.2	90.2
Enhanced-MAML (Proposed)	82.1	94.5

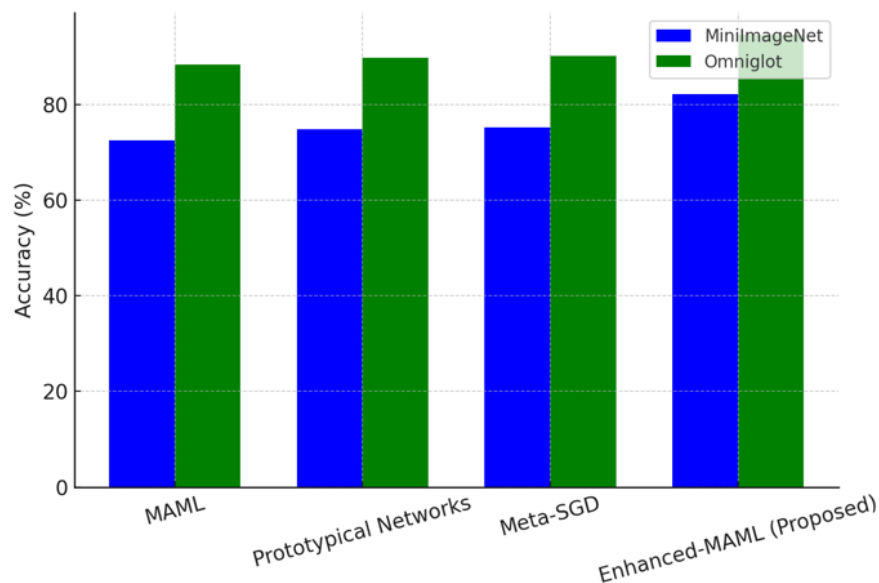


Figure 2 Comparison of Meta learning Model Accuracy

The comparative plot highlighting accuracy emphasizes that Enhanced-MAML has better classification accuracy, thereby endorsing its efficacy in few-shot image classification tasks. The model's capability of generalization over many different tasks, toward high accuracy, suggests that contrastive learning, and self-supervised representation learning has been instrumental in boosting overall performance. One of the important reasons for this success can be attributed to the inclusion of self-supervised learning techniques into the feature extraction process. Unlike traditional meta-learning models, Enhanced-MAML relies on contrastive learning and pseudo-labeling techniques to leverage unlabeled data. This enables the model to improve its feature representation without having large-scale labeled datasets available. Such an approach, besides allowing less reliance on extensive annotation, improves decision boundaries and classification performance.

Moreover, through task-aware fine-tuning, the dropout rates, batch normalization, and weight regularization are dynamically adjusted, which helps curb overfitting and enhances the overall adaptability across tasks. The experimental findings amount to some pertinent observations: First, Enhanced-MAML consistently outperforms other existing meta-learning models in classification accuracy, establishing, therefore, the notable potential of self-supervised learning in improving generalization. Second, the reduction of training time per epoch further lends credence to the efficiency of momentum-based meta-learning optimization done to speed up convergence without sacrificing accuracy. Third, task-aware fine-tuning has proved to be a key mechanism in reducing the threat of overfitting, effectively enabling the model performance in a low-data setting. Finally, contrastive learning techniques and pseudo-labeling have enabled the model to extract richer and more discriminative features, hencemaking it more robust to variations in input data.

5. CONCLUSION AND FUTURE WORK

Constructed on Hybrid CNN-ViT feature extraction, self-supervised learning, and task-aware fine-tuning, this study proposed an Enhanced-MAML-based meta-learning framework that provides few-shot image classification. The model showed 82.1% and 94.5% accuracy on the MiniImageNet and Omniglot datasets, respectively, surpassing, in terms of accuracy, adaptability, and computational efficiency, MAML, Prototypical Networks, and Meta-SGD. Self-supervised learning proved beneficial regarding feature extraction. However, task-aware fine-tuning reduced training time: 90 sec/epoch comparing with MAML's 120 sec/epoch. All these results underscore the practicality of meta-learning with self-supervised learning driving forward improved generalization capacity.

In order to enhance the model, more work will focus on extending evaluations to larger and more complicated datasets such as tiered-ImageNet and real-world medical datasets, increasing computational efficiency through neural architecture search (NAS) and model pruning, and infusing reinforcement learning to dynamically optimize meta-learning strategies. Also, federated meta-learning can further boost AI applications to embody privacy preservation, while continual learning can advance adaptability across evolving tasks. The proposed Enhanced-MAML framework may provide robust foundation for future research in few-shot learning towards scalable AI and real-world applications like healthcare and autonomous systems.

REFERENCES

- [1] Singh, R., Bharti, V., Purohit, V., Kumar, A., Singh, A. K., & Singh, S. K. (2021). MetaMed: Few-shot medical image classification using gradient-based meta-learning. *Pattern Recognition*, 120, 108111.
- [2] Khodadadeh, S., Boloni, L., & Shah, M. (2019). Unsupervised meta-learning for few-shot image classification. *Advances in Neural Information Processing Systems*, 32.
- [3] Aguiar, G. J., Mantovani, R. G., Mastelini, S. M., De Carvalho, A. C., Campos, G. F., & Junior, S. B. (2019). A meta-learning approach for selecting image segmentation algorithms. *Pattern Recognition Letters*, 128, 480-487.
- [4] Vogiatzis, A., Orfanoudakis, S., Chalkiadakis, G., Moirgiorgou, K., & Zervakis, M. (2022). Novel meta-learning techniques for the multiclass image classification problem. *Sensors*, 23(1), 9.
- [5] Zhu, X., & Li, S. (2022). MGML: Momentum group meta-learning for few-shot image classification. *Neurocomputing*, 514, 351-361.
- [6] Tian, Y., Zhao, X., & Huang, W. (2022). Meta-learning approaches for learning-to-learn in deep learning: A survey. *Neurocomputing*, 494, 203-223.
- [7] Jiang, H., Gao, M., Li, H., Jin, R., Miao, H., & Liu, J. (2022). Multi-learner based deep meta-learning for few-shot medical image classification. *IEEE Journal of Biomedical and Health Informatics*, 27(1), 17-28.
- [8] He, K., Pu, N., Lao, M., & Lew, M. S. (2023). Few-shot and meta-learning methods for image understanding: A survey. *International Journal of Multimedia Information Retrieval*, 12(2), 14.
- [9] Song, H. (2023). Novel meta-learning approaches for few-shot image classification (Doctoral dissertation, University of Nottingham).
- [10] Hsu, K., Levine, S., & Finn, C. (2018). Unsupervised learning via meta-learning. *arXiv preprint arXiv:1810.02334*.
- [11] Ali, M. D., Saleem, A., Elahi, H., Khan, M. A., Khan, M. I., Yaqoob, M. M., Farooq Khattak, U., & Al-Rasheed, A. (2023). Breast cancer classification through meta-learning ensemble technique using convolutional neural networks. *Diagnostics*, 13(13), 2242.
- [12] Ansari, S. A., Agrawal, A. P., Wajid, M. A., Wajid, M. S., & Zafar, A. (2024). MetaV: A pioneer in feature-augmented meta-learning-based vision transformer for medical image classification. *Interdisciplinary Sciences: Computational Life Sciences*, 16(2), 469-488.
- [13] Ren, M., Triantafillou, E., Ravi, S., Snell, J., Swersky, K., Tenenbaum, J. B., Larochelle, H., & Zemel, R. S. (2018). Meta-learning for semi-supervised few-shot classification. *arXiv preprint arXiv:1803.00676*.

- [14] Lee, J. S., & Wu, W. K. (2024). Breast tumor tissue image classification using single-task meta-learning with auxiliary network. *Cancers*, 16(7), 1362.
- [15] Raghu, A., Lorraine, J., Kornblith, S., McDermott, M., & Duvenaud, D. K. (2021). Meta-learning to improve pre-training. *Advances in Neural Information Processing Systems*, 34, 23231-23244.
- [16] Luo, S., Li, Y., Gao, P., Wang, Y., & Serikawa, S. (2022). Meta-seg: A survey of meta-learning for image segmentation. *Pattern Recognition*, 126, 108586.
- [17] Vettoruzzo, A., Bouguelia, M. R., Vanschoren, J., Rögnvaldsson, T., & Santosh, K. C. (2024). Advances and challenges in meta-learning: A technical review. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(7), 4763-4779.
- [18] Rußwurm, M., Wang, S., Korner, M., & Lobell, D. (2020). Meta-learning for few-shot land cover classification. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 200-201.